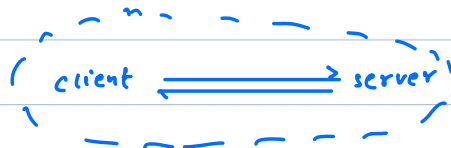


HTTP connections



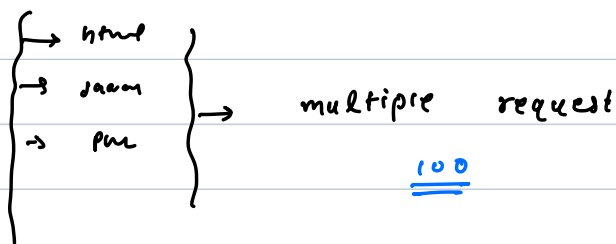
HTTP 1.0 :-

non persistent connection.

each connection could only handle one request at a time.

• request are res.

facebook.com



HTTP 1.1

persistent connection (keep-alive)

create connection

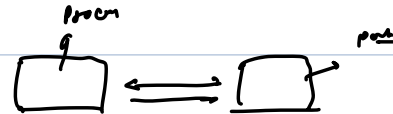
req 1 res 1
req 2 :
: :

connection end

port numbers

it is an end point of network.

transport layer

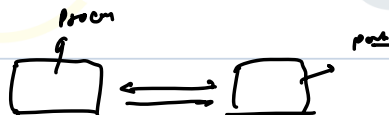


socket address

ip address + port number

it is combination of ip address and port number.

transport layer



tracking

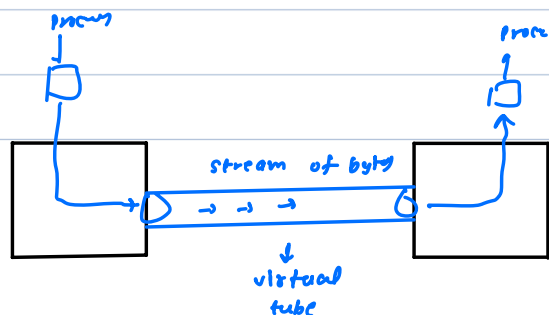
guarantee, non guarantee

ordering

TCP and UDP

transmission control protocol.

it established a virtual tube.



• it is reliable

acknowledgement

it is ensure that data is reached at receiver side.

sender

Receiver



if packet is loss then it retransmitted the segment

sender

1000 bytes

large file

receive

[800 bytes]

partial part

In transport layer data is called as segments

(20-60) bytes

TH

data



segment

- source port number
- destination port number
- sequence number

1 2 3 4 5 6

sender

4 3 1 2

Receiver

$\begin{array}{ccc} \text{01} & \text{02} & \text{03} \\ \hline 1-99 & 100-199 & \dots \end{array}$

At receiver end we collect the bytes and rearrange.

- **acknowledge number** : indicate next byte of the data to be send.

$1999 \longrightarrow \text{acknowledge } 2000$

Header length (4 bit)

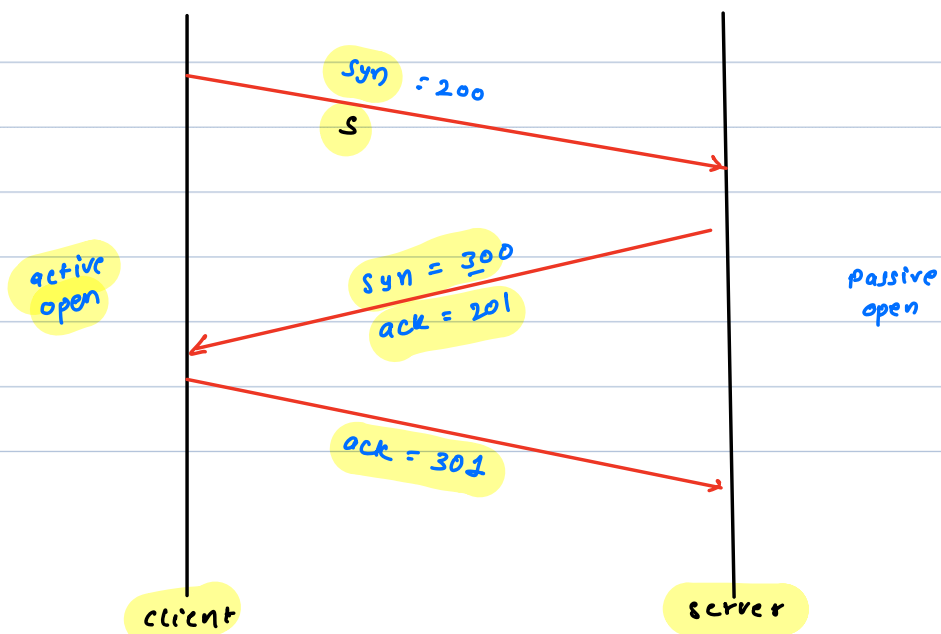
$$5 \times 4 = 20$$

$$7 \times 4 = 28$$

$$15 \times 4 = 60$$

$$(20 - 60)$$

- **TCP connection establishment**: the process is known as **three way hand shaking**.



* error handling

checksum is used for error handling

it is used to find an error like missing of a byte.

eg: $\left(\begin{array}{c} 8 \text{ bit} \\ 11100110 \end{array} \right) \begin{array}{c} 8 \text{ bit} \\ 11001010 \end{array} \begin{array}{c} 8 \text{ bit} \\ 10110101 \end{array}$

10 111111
11100110
11001010
10110101

10 01100101
10

01100111

10011000 → check sum

wrap up if
value is exceed

At receiver end

8 bit 8 bit 8 bit
(1 1 1 0 0 1 1 0) 1 1 0 0 1 0 1 0 | 1 0 1 1 0 1 0 1)

10 1 1 1 1 1 1 0
1 0 0 1 1 0 0 0
1 1 1 0 0 1 1 0
1 1 0 0 1 0 1 0
1 0 1 1 0 1 0 1

10 1 1 1 1 1 1 0 1

→ wrap up

1 1 1 1 1 1 1 1

if every bit is on it means there
is no error.

10 1 1
10 1 0 1 0 1
1 0 0 1 1 0 0 0

1 1 1 0 0 1 1 0

1 1 0 0 1 1 1 0

1 0 1 1 0 1 0 1

0 0 0 0 0 0 0 1

0 0

0 0 0 0 1 0 1

→ error

if every bit is not on it means there is an error.

* flow control



• window size



it is connection oriented.

it is slow.

UDP = user datagram protocol.

1) connectionless

each datagram is independent.

2) unreliable delivery

udp does not guarantee delivery of packet.

3) fast transmission:

4) no ordering guarantee

online gaming → udp

